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Full Length Article

# Multi-Sized porous silica/PDMS composite layer for enhanced performance in triboelectric nanogenerators☆

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ABSTRACT

This study explores the enhancement of triboelectric nanogenerator (TENG) performance through the incorporation of a KIT-6/PDMS composite friction layer featuring multi-sized pores. KIT-6, a mesoporous silica with a well-ordered structure, was integrated with polydimethylsiloxane (PDMS) to create a hybrid material that leverages the synergistic effects of both components. The nano-sized pores in the KIT-6 framework significantly increase the surface area and provide diverse charge trapping sites, which are crucial for improving triboelectric performance. The KIT-6/PDMS composite demonstrated a remarkable increase in performances compared to conventional PDMS-based TENGs. The enhancement is due to the distinctive porous structure that facilitates more effective contact electrification and charge separation. The TENG with 2 wt% KIT-6/PDMS achieved an instantaneous power density of 4.37 W/m$^2$, a 6.52-fold improvement over the TENG with pure PDMS. Additionally, the energy produced by the KIT-6/PDMS nanogenerator is accumulated in a circuit and subsequently utilized to light up 84 LEDs. The study highlights the potential of porous nanocomposites in advancing the efficiency of energy harvesting devices, laying the foundation for more sustainable and efficient power sources in wearable electronics and IoT applications. The findings underscore the importance of material innovation in optimizing the functionality of next generation nanogenerators.

## 1. Introduction

The increasing energy costs pose one of the most pressing challenges to human life, productivity, and mobility [1]. Moreover, projections indicate that the world's recoverable fossil fuel reserves may be depleted by the end of this century, further emphasizing the urgency of transitioning to sustainable energy solutions [2]. Energy transition has become a highly important topic, particularly due to the overreliance on conventional energy sources like coal, coupled with limited exploitation of renewable energy alternatives under specific natural, technical, economic, and social circumstances [3]. In nature, various forms of ambient energy abound, including wind, low-frequency vibrations, body motion, tides, raindrops, and waves, etc. These energy sources are typically low-frequency, tiny, random and microscale, are often neglected and wasted because of the lack of effective methods to capture them. A TENG is an energy harvesting device that utilizes the triboelectric effect to convert mechanical energy, such as motion or vibration, into electrical energy. It operates by utilizing two dissimilar materials that become electrically charged upon contact and subsequent separation [4]. This process generates a small electric current that can be utilized to operate small devices. The efficiency and reliability of such TENGs largely depend on the properties of the friction layer, which facilitates charge transfer and significantly enhances the device's overall performance [5].

In recent years, there has been significant attention directed toward enhancing both the materials and structure of friction layers to boost TENG efficiency [6,7]. As an illustration, Nardekar and colleagues utilized the unique dual-functional properties of exfoliated $MoS_2$ sheets in self charging power system by creating micro-supercapacitors integrated with a TENG system [8]. They used $MoS_2$ sheets as a charge trapping interfacial layer, which improved the energy harvesting efficiency. This approach demonstrates the potential of integrating advanced nanomaterials into TENGs to improve their performance and multifunctionality. Among the various materials being explored for friction layers, PDMS stands out for its excellent chemical stability,

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mechanical flexibility, cost effectiveness, transparency, and tunable surface properties [9–11]. PDMS ranks as one of the most negatively charged triboelectric materials, coming just after poly(tetrafluoroethylene) (PTFE). However, PDMS is in much higher demand than PTFE due to its transparent and flexible properties. Additionally, PDMS has the ability to combine with different nanostructured materials, resulting in the formation of composite thin films [12,13]. Moreover, the coating ability of PDMS on other surfaces is superior to that of PTFE. A critical aspect is that, for both self-powered sensing and energy harvesting, it is essential to continuously optimize the TENG's output performance to guarantee practical usability. The surface morphology is crucial for enhancing electrification by enlarging the contact area. Several experiments have been done that altering the surface morphology or internal structure of nanogenerating layers can greatly enhance the output efficiency of TENGs. For instance, Chun and colleagues showed that mesoporous PDMS films embedded with Au nanoparticles greatly enhance the efficiency of nanogenerators operating in vertical-contact separation mode [14]. Chen et al. studied a composite porous PDMS film-based TENG that combines high dielectricity and porosity, demonstrating that charge transfer and surface charge density are strongly correlated with the porosity and relative permittivity of the tribo-material [15]. Lastly, Yang and co-researchers designed a textile-based TENG by incorporating commercial conductive fabric with a porous PDMS embedded with graphene oxide [16].

The output efficiency of TENGs is closely linked to the surface charge density of the electrostatic layer, which depends on the inherent contact-induced electrostatic properties of the triboelectric materials. In a given environment, the charge transfer capacity between two specific friction materials is capped at a certain limit. Although modifying the inner structure or surface can effectively enhance the performance, surface charges have a tendency to dissipate rapidly, particularly in high-moisture and high-temperature conditions. Consequently, the output performance of structured TENGs will gradually decrease as the enhanced surface charge density declines. Thus, it is crucial to prevent this situation. Previous studies have shown that internal body charges within triboelectric materials dissipate much more slowly than surface charges and can persist for extended periods of time [16–18]. Additionally, when exposed to a strong electric field and a steep charge concentration gradient, a significant number of surface charges are able to migrate into the inner regions of the triboelectric materials through diffusion and charge drift [14]. Without effective storage sites or charge traps inside the triboelectric material these internalized charges will ultimately recombine with the charges of opposite sign on the electrodes or disperse into the surrounding environment, thereby reducing output performance. To address this issue, integrating nanomaterials with high electron trapping or storage capabilities into triboelectric materials has been shown to effectively in retaining charges and enhancing overall charge density. The effectiveness of charge trapping or storage within the material is influenced not only by its intrinsic electron-attracting characteristics but also by the specific surface area (SSA) of the nanomaterials used $SiO_2$ is environmentally friendly and non-toxic, making it a sustainable choice for the additive material. This property aligns with the growing emphasis on environmentally conscious technologies. For example, Li et al. suggested employing well-structured mesoporous silica particles with a high surface area to serve as efficient storage sites for charges within PDMS, aiming to improve the performance of devices. [18]. Incorporating $SiO_2$ can alter the surface texture of the friction layer, resulting in increased roughness. This increased roughness can enhance the contact surface area between surfaces, resulting in more effective charges generation during triboelectric interactions. As a hard and durable material, $SiO_2$ can increase the mechanical strength of the friction layer. This leads to greater wear resistance and longevity, which are critical for the consistent performance of TENG devices.

Among all kinds of $SiO_2$ nanomaterials, mesoporous KIT-6 is a type of *meso*-porous silica characterized by its unique cubic structure, typically exhibiting the *Ia3d* symmetry. The structure is recognized for its large surface area, considerable pore capacity, and uniform pore dimensions, making it beneficial for various applications such as adsorption, catalysis, and drug delivery [19–22]. In contrast, conventional MCM-41 and MCM-48 suffer from structural collapse and degradation of physicochemical properties under high pelletizing pressure and humid conditions due to their thinner pore walls [23]. KIT-6, a bicontinuous cubic mesoporous material with an *Ia3d* structure, provides a high specific surface area and large pore sizes, both of which are essential for enhancing TENG performance [24]. However, up to now, there are few papers reporting the introduction of KIT-6 into PDMS and using the KIT-6/PDMS composite as the friction layer to improve TENG performance. Additionally, there are no reports that combine *meso*-porous KIT-6 with porous PDMS to enhance TENG output performance using a dual-porous material strategy.

This study presents a novel approach to improve TENG performance by integrating a KIT-6/PDMS friction layer with varying pore sizes. In this paper, we focus on evaluating how the integration of KIT-6 with PDMS improves TENG performance, investigating the mechanisms underlying this improvement, and exploring the influence of surface area and pore structure on charge storage and transport. The use of non-halide, cost-effective, easy-to-fabricate, and non-toxic *meso*-porous KIT-6, which demonstrates similar or better performance, adds to the novelty of this work. The porous structure of PDMS has been shown to offer greater flexibility compared to a flat film, and introducing internal micro-porous structures within PDMS can notably enhance the efficiency of TENGs [14,25]. Additionally, mesoporous KIT-6 particles can serve as effective storage sites for body charges within the PDMS film, further enhancing TENG performance. The utilization of KIT-6 with its well-defined porous structure offers increased surface area and improved charge trapping capabilities, while PDMS provides mechanical robustness and flexibility. The effects of the concentrations of KIT-6, applied force along with operation frequency on the performance of the TENGs have been comprehensively examined. Our findings demonstrate that the integrating pores into the friction layer and adding KIT-6 notably enhances the triboelectric performance of TENGs, leading to improved power generation capabilities and potential applications in energy harvesting systems, self-powered sensors, wearable electronics. Overall, this study offers valuable insights into the optimization and design of friction layer materials in TENGs, advancing the development of sustainable energy technologies.

## 2. Experiment section

### 2.1. Materials

The P123 (poly(ethylene glycol)-block-poly(propylene glycol)-block-poly(ethylene glycol)), TEOS (tetraethyl orthosilicate), 1-butanol, and ethanol were all obtained from Sigma-Aldrich. HCl and $NH_3{\cdot}H_2O$ were sourced from the Daejung chemical &metals company. Copper was obtained from Ducksung Hitech. The 184 silicone elastomer base and the 184 silicone elastomer hardener were acquired from Dow Chemical Company. All reagents were utilized as received.

### 2.2. Fabrication of the device

#### 2.2.1. Synthesize of KIT-6

KIT-6 was synthesized as follows. Specifically, 2 g of P123 and 3.3 mL of HCl (~35 wt%) were dissolved in 80.5 g of $H_2O$. The solution was stirred thoroughly overnight at ambient temperature to achieve a clear and uniform mixture, free of any P123 lumps. The solution was then stirred at a moderate speed and heated to 37 °C. Next, 2 g of n-butanol were slowly incorporated into the mixture. After 1.5 h, 4.3 g of TEOS were added, and the solution was stirred at 37 °C for 25 h. The resulting mixture was then transferred to a blue-capped bottle and heated in an oven at 120 °C for a minimum of 26 h. The solution was allowed to cool slowly to ambient temperature. The resulting white product was rinsed

three times with purified water and ethanol, then filtered to remove the solvent. It was dried at 80 °C overnight and subsequently calcined at 540 °C for 6 h to yield KIT-6.

#### 2.2.2. Synthesize of $SiO_2$ nanoparticles

The non-mesoporous $SiO_2$ nanomaterials were synthesized using a previously reported method with minor modifications. First, 2.6 g of TEOS and 5.75 g of ethanol were mixed in a glass bottle to form Solution A. At the same time, 3.05 g of $NH_3{\cdot}H_2O$, 2.3 g of ethanol, and 2.25 g of water were combined and stirred to form a uniform solution called Solution B. Solutions A and B were combined in one bottle and mixed thoroughly for 1.5 h. The mixture was then dried at 100 °C overnight and calcined at 125 °C for 26 h or so, resulting in the formation of $SiO_2$ nanoparticles.

#### 2.2.3. Assemble/fabrication of TENGs

The fabrication process followed our previous report. The PDMS base and PDMS hardener, mixed in a 10:1 wt ratio, were combined in a bottle and stirred for several minutes. Then, different ratios of KIT-6/$SiO_2$ particles were mixed to the PDMS solution and stirred. The mixture needed to be then vacuumed for more than 30 min to remove any bubbles. Next, a specified amount of $H_2O$ was then added and stirred for 30 min to achieve a slurry. This slurry was then applied to a copper substrate using a blade coating technique, resulting in a PDMS layer approximately 100 μm thick. The thin film was dried at 90 °C for 2 h. In our experiment, another friction layer was pure copper, and the entire TENG device was assembled. For the non-porous PDMS-based TENG, the process was the same except for the addition of water, which was omitted.

### 2.3. Characterization

For the detailed analysis, various instruments such as FE-SEM, X-ray diffraction, a stylus-based profiling system, a high-speed camera, a porosimeter, an oscilloscope, and an electrometer were used. Further details are provided in the supporting information.

## 3. Results

The ordered mesoporous structure of KIT-6, characterized by its Ia3d symmetry, was confirmed through small-angle XRD patterns (Fig. 1a). Compared with the XRD data of ordinary $SiO_2$ nanomaterials (depicted in Fig. S1c), this pattern of KIT-6 revealed four distinct peaks, corresponding to (211), (220), and (332) planes, respectively. The (211) plane indicated a high degree of ordering within the mesoporous structure, while the (220) plane suggested a well-defined cubic arrangement. The (332) peak reinforced the cubic symmetry. These peaks indicated an ordered mesoporous structure in KIT-6, consistent

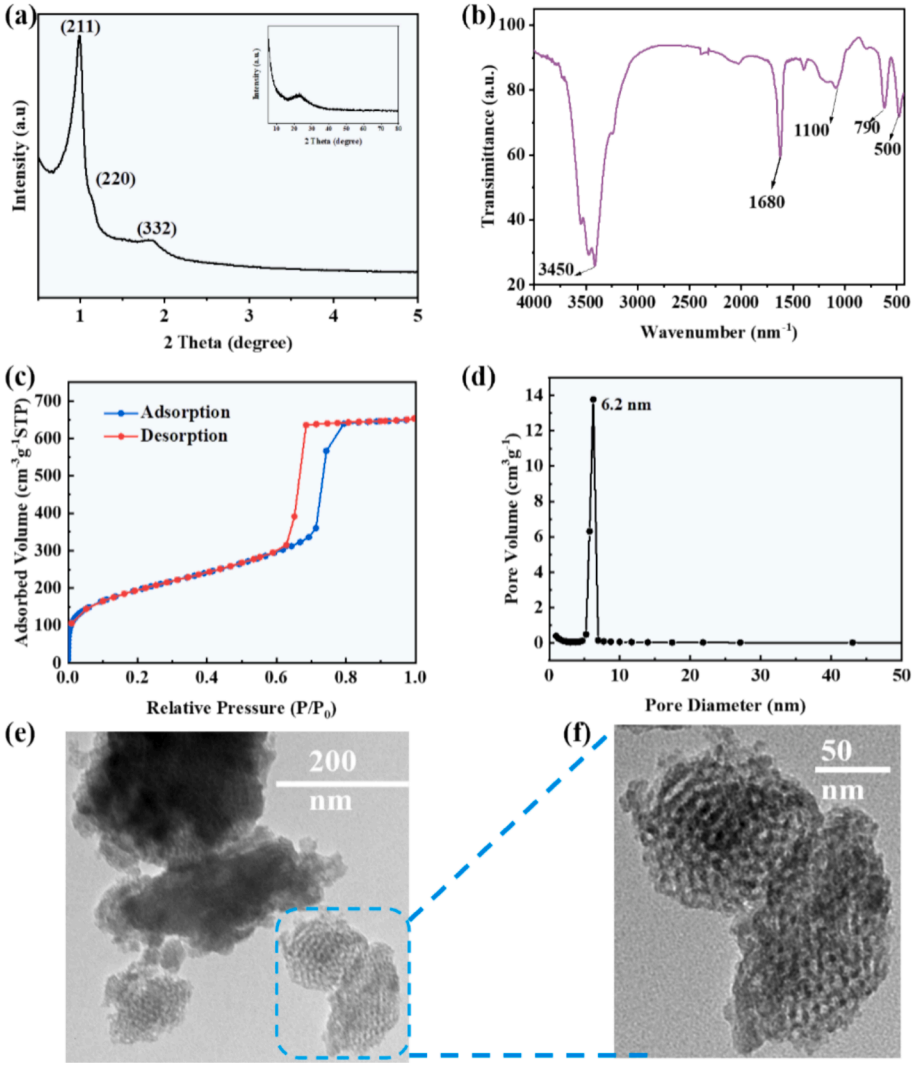



**Fig. 1.** Various results of KIT-6 (a) XRD curves; (b) FTIR data; (c) $N_2$ adsorption–desorption isotherms; (d) Pore size distribution curve; (e, f) TEM micrographics.

with previous reports [26,27].

Additionally, a wide-angle XRD test on the KIT-6 powder material (inset of Fig. 1a) showed a single, broad, low diffraction peak at $2\theta = 22^{\circ}$, suggesting that silica lacks a fixed crystalline structure [28,29]. The infrared spectrum of KIT-6 matched previous reports, as shown in Fig. 1b, and no organic absorption peaks were observed, confirming that the organic components in the raw materials were completely removed during calcination at 550 ℃, indicating high purity in the synthesized KIT-6 [30,31].

Fig. 1c showed the $N_2$ adsorption–desorption isotherms of KIT-6. H1-type hysteresis loops at pressures ranging from approximately 0.6 to 0.8 confirm the presence of mesoporous cylindrical channel structures. The pore size distributions, calculated from the desorption branch, are centered at 6.2 nm (Fig. 1d). The SSA of KIT-6 was 675 $m^2/g$. Additionally, the morphology of KIT-6 was observed using TEM. Fig. 1e, f demonstrated the successful formation of the KIT-6 structure. The centers of two adjacent pores were approximately 8 nm apart, and the diameter of pore was 6.2 nm. All characterizations confirmed that the synthesized KIT-6 exhibits high purity, a significant surface area and a consistent pore size distribution [29,32].

The working mechanism of the fabricated TENG is illustrated in Fig. 2. In the initial state (Fig. 2-i), no charge generation or accumulation occurs. When an external force causes contact between the two friction layers (the Cu layer and the KIT-6/porous PDMS layer), electrons transfer from the Cu layer to the KIT-6/porous PDMS layer due to differences in electron affinity. Porous PDMS tends to gain electrons, while copper tends to lose electrons. When the two surfaces come into contact, a potential difference arises, leading to the transfer of electrons from the Cu layer to the porous PDMS layer. The next step involves separating the two friction layers (Fig. 2-ii). When the external pressure isn't applied, the two friction layers are pulled apart. This separation induces a potential difference through electrostatic induction. As the separation between the two layers grows, the potential increases, causing electrons to move from the electrode of the KIT-6/porous PDMS film to the electrode of the copper film, thus producing a downward current. After releasing step, the device is reaching to equilibrium state (Fig. 2-iii). When the external force is completely withdrawn, the separation between the copper layer and the KIT-6/porous PDMS film reaches its peak. At this equilibrium point, no current is generated. Final step is a recontact step (Fig. 2-iv). When the external force is reapplied and the layers are brought closer together, the potential decreases, resulting in an upward current. Repeated application and removal of the external force generate a consistent alternating current (AC) signal in the circuit. Throughout the contact-separation cycle, electrons from the surface can infiltrate the porous PDMS for the following reasons: drift due to the strong electric field and diffusion stemming from differences in electron concentration. The mesoporous KIT-6 nanoparticles within the PDMS capture and store these electrons until reaching saturation. Once saturation is achieved, the TENG operates steadily, ensuring uniform charge transfer during each cycle.

The PDMS-based TENG was mounted on a 4.5 cm stroke stopper. The device was moved at a frequency of 2 Hz. The performance tests were conducted applying 5 N to the device. The experimental results, depicted in Fig. 3a–c, show data for pristine PDMS, porous PDMS, and 0.5 wt % KIT-6@porous PDMS. The corresponding voltage, short-circuit current, and charge transfer density for these samples were 86 V, 7.5 μA, and 33.8 $\mu C/m^2$; 116 V, 10 μA, and 46 $\mu C/m^2$; and 135 V, 12.76 μA, and 55.4 $\mu C/m^2$, respectively. These results indicate which the performance of this nanogenerator, including voltage, current, and charge transfer density, improves when transitioning from a pristine (flat) structure to a porous structure. Compared to the unmodified TENG, the output voltage, current, and charge density increased approximately 1.35, 1.33, and 1.36 times, respectively. This finding suggests that introducing porosity into PDMS is an efficient approach to enhancing the electrical output performance of PDMS-based TENGs. It also highlights the critical role that pore structure plays in influencing the performance of these devices. When a device with porous layer is compressed by an external force, the film is reduced to its minimum thickness, resulting in an increase in output signals due to the corresponding increase in capacitance [15]. Additionally, when a small amount of mesoporous KIT-6 (0.5 wt%) was incorporated into the porous PDMS, the electrical output of the device was further enhanced compared to the TENG without KIT-6. The large specific surface area of KIT-6 increases the contact area with PDMS, facilitating charge transfer and storage. In addition, the porous structure of KIT-6 allows it to act as a charge trap, effectively capturing and retaining charges generated during operation. For the device with KIT-6/porous PDMS layer, the voltage, current, and charge transfer density improved by approximately 1.57, 1.7, and 1.64 times, respectively, in comparison to the original PDMS TENG. This result demonstrates that the incorporation of mesoporous KIT-6 can improve the output of nanogenerators. Specifically, the dispersed KIT-6 ($SiO_2$) within the PDMS thin film served as charge trapping sites. This enhanced the available interface for charge storage. As a result, more charges were captured around the $SiO_2$ nanoparticles. This effect is illustrated in the inset of Fig. 2-ii [33,34]. Secondly, the addition of $SiO_2$ can modify the surface texture of the friction layer, increasing its roughness. This increased roughness improves the contact area between surfaces, resulting in more effective charge generation during triboelectric interactions [35].

We found that adding 0.5 wt% KIT-6 to porous PDMS significantly improved the output of nanogenerator. To explore the relationship between the KIT-6 additive amount and the electrical performance of TENGs, we conducted experiments with porous PDMS containing different ratios of KIT-6, comparing, and analyzing their performance. Fig. 3d–f show the output voltage, short circuit current, and charge transfer density of the KIT-6/porous PDMS TENG as a function of KIT-6

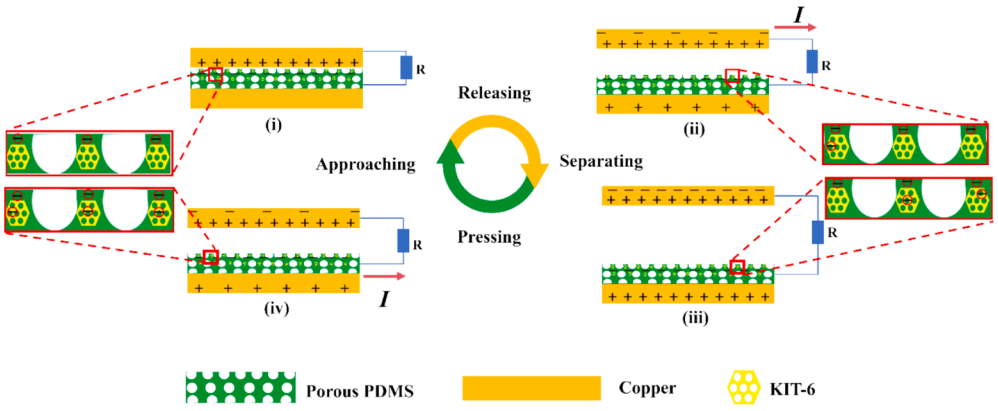



**Fig. 2.** Functional mechanism of a KIT-6/porous PDMS based TENG.

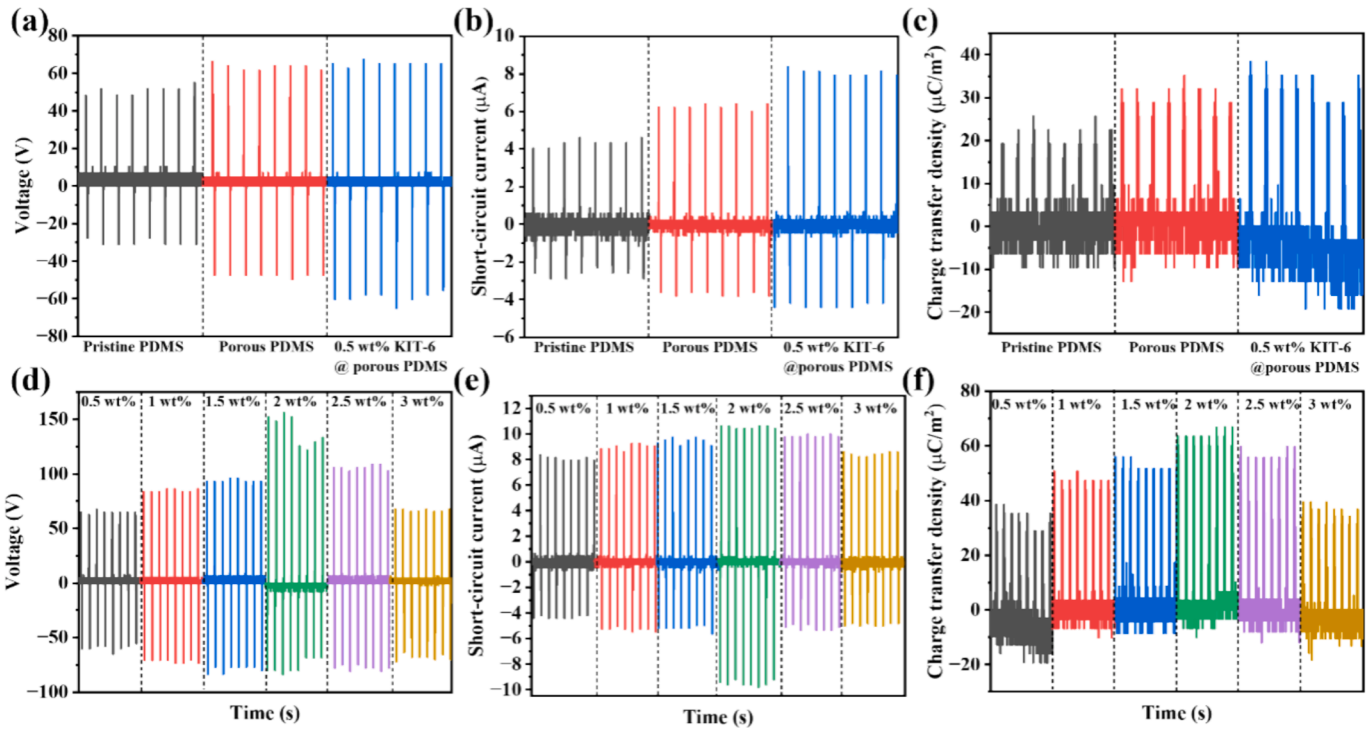



**Fig. 3.** Comparison of (a) output voltage, (b) current, and (c) charge transfer density for TENGs with different friction surfaces. Additionally, (d) output voltage, (e) current, and (f) charge transfer density are compared for nanogenerators with different concentrations of mesoporous KIT-6.

concentration ranging from 0.5 to 3 wt%. Based on our experimental results, when the KIT-6 ratio was below 2 wt%, the voltage, current, and charge transfer density improved as the KIT-6 ratio increased. This is because the mesoporous KIT-6 nanoparticles serve as efficient charge storage sites, thereby enhancing TENG performances. When this ratio reached 2 wt%, the output performance peaked at 254 V, 20.4 μA, and 70.8 μC/m$^2$ for voltage, current, and charge transfer density, respectively. The decrease in performance when the KIT-6 concentration is greater than 2 wt% is attributed to a reduction in the effective electrified PDMS region as the KIT-6 nanoparticles tend to accumulate on the surface [18,34]. Consequently, the rise in charge storage sites, combined with the reduction in the PDMS area at higher KIT-6 concentrations, leads to a decline in TENG performance. First, at higher concentrations, KIT-6 particles tend to aggregate on the PDMS surface, reducing the effective electrified area. Second, excessive KIT-6 loading can alter the mechanical flexibility of the PDMS, potentially affecting the contact separation process critical to triboelectric power generation. These highlight the importance of optimizing the concentration of KIT-6 to balance the benefits of increased charge storage with the need to maintain sufficient PDMS surface area for effective triboelectric interactions. These results demonstrate that the incorporation of KIT-6 can effectively improve the efficiency of PDMS-based nanogenerators. Furthermore, the ratio of KIT-6 is crucial in determining the overall performance of the device.

In addition, we also tested the output performance of TENG using another friction layer together with KIT-6/porous PDMS. In this instance, a teflon film was selected as another friction layer. As illustrated in Fig. S2a–c, the TENG with a teflon layer and 0.5 wt% KIT-6/porous PDMS membrane achieved a peak output voltage of 140 V, a short-circuit current of 12.4 μA, and a charge transfer density of 62 μC/m$^2$, outperforming both pure PDMS and porous PDMS (2 Hz and 5 N). When 2 wt% KIT-6 was incorporated into the porous PDMS membrane under the same conditions, the output voltage increased to 260 V (Fig. S2d). Additionally, the short-circuit current rose significantly from 12.4 μA to 21.2 μA with the addition of 2 wt% mesoporous KIT-6 powder, as displayed in Fig. S2e. The charge transfer density also improved from 62 μC/m$^2$ to 93 μC/m$^2$ when the addition of KIT-6 reached 2 wt%. However, a comparable decline in electrical output was observed when the ratio of KIT-6 exceeded 2 wt%. This experiment clearly demonstrates the positive impact of integrating KIT-6 into the porous PDMS matrix. The significant improvements in voltage, current, and charge transfer density underscore the effectiveness of mesoporous KIT-6 in improving the triboelectric performance of nanogenerators. However, the observed degradation beyond the 2 wt% concentration threshold indicates an optimal KIT-6 concentration for maximizing the output performance.

We conducted a series of characterizations on KIT-6/porous PDMS to better understand its role in improving TENG performance. Wide-angle XRD data in Fig. 4a shows no significant differences among pristine PDMS, porous PDMS, and KIT-6/porous PDMS, suggesting that structural changes do not affect the PDMS itself. Furthermore, after introducing 2 wt% KIT-6, no peaks related to $SiO_2$ were observed, indicating that the quantity of KIT-6 added was too small to be detected. Given KIT-6′s hydrophilic nature, we also examined the hydrophobicity of these three materials to understand whether adding KIT-6 affects the hydrophobicity of PDMS, potentially impacting the TENG's performance. Water contact angle measurements in Fig. 4b–d showed similar angles for pristine PDMS, porous PDMS, and KIT-6@porous PDMS, indicating that adding a small amount of KIT-6 does not affect the hydrophobicity of PDMS. Surface roughness is an important factor in increasing the contact area of TENG devices, as changes in the roughness of the friction layer can alter the electrostatic potential and field, impacting capacitance. Fig. 4e summarizes the measured average surface roughness (Ra) for these materials. A low Ra of 8 μm was found for pure PDMS. After transitioning from a flat structure to a porous one, Ra increased to 35 μm. The addition of 2 wt% KIT-6 further increased Ra to 126 μm. This increase in roughness explains the significant improvement in TENG output performance after introducing KIT-6, aligning with previous research indicating that output voltage and capacitance increase with surface roughness [35,36]. Finally, we observed the morphology of KIT-6/porous PDMS using scanning electron microscopy (SEM). Fig. 4f–g shows many pores on the surface with several KIT-6 nanoparticles

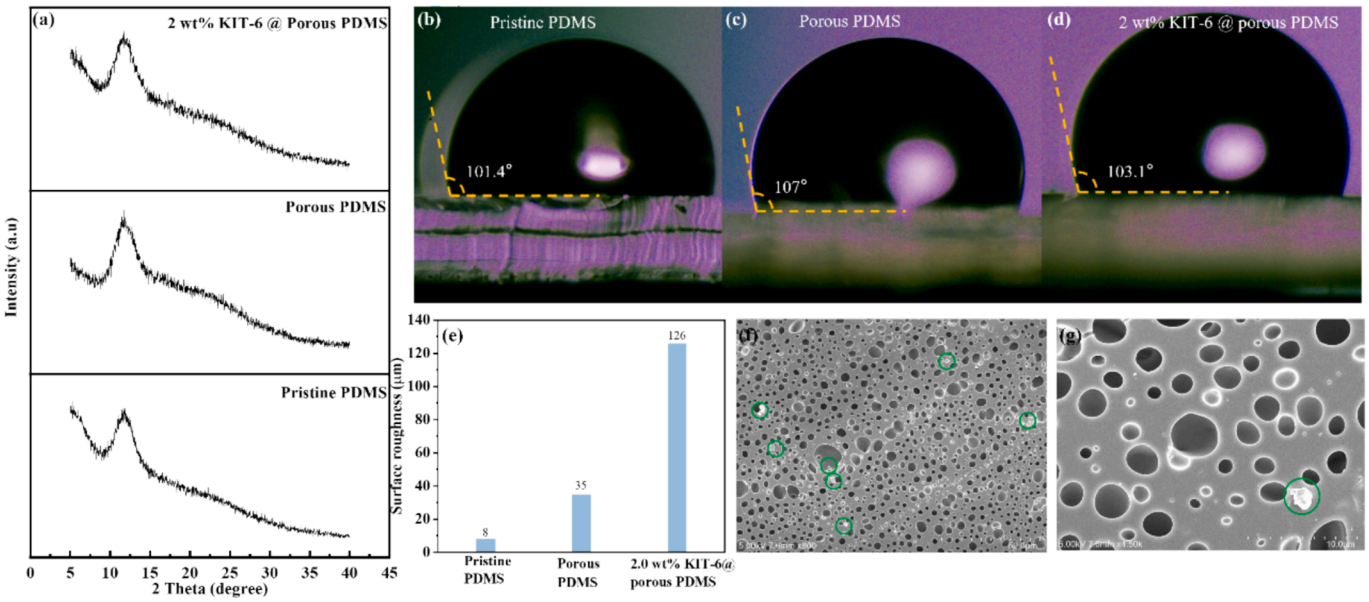



**Fig. 4.** (a) XRD spectrums of pristine PDMS, porous PDMS, and 2 wt% KIT-6@porous PDMS. Water contact angles of (b) pristine PDMS, (c) porous PDMS, and (d) 2 wt% KIT-6@porous PDMS. (e) Average surface roughness of pristine PDMS, porous PDMS, and 2 wt% KIT-6@porous PDMS. (f) and (g) SEM micrographics of 2 wt% KIT-6@porous PDMS.

dispersed within the porous structure. The porous design of the PDMS notably enhances the contact area in TENGs. Furthermore, the randomly distributed KIT-6 within the porous PDMS film acts as a charge storage site, boosting the overall performance of the nanogenerator.

Although our experimental results showed that both porous PDMS structures and KIT-6 contribute to improved TENG performance, it is important to determine which factor plays a more significant role. First, we evaluated the electrical performance of TENGs composed of porous PDMS with KIT-6 against those made from flat PDMS and KIT-6. According to the results (Fig. 5a, b), TENG devices with porous PDMS exhibited better performance than those with flat PDMS and KIT-6. Additionally, when the ratio of KIT-6 reached 2 wt%, the difference in output voltage and current between the two devices was at its peak. This finding indicates that the porous structure is a critical factor in improving TENG performance. Next, to investigate the role about the mesoporous structure in $SiO_2$ in enhancing TENG performance, we synthesized non-porous $SiO_2$ and incorporated it into porous PDMS, then assembled TENG devices under the same conditions. Fig. 5a, b and Table 1 represent the output voltage and short-circuit current for KIT-6-based TENGs and $SiO_2$-based TENGs, respectively. As the ratio of KIT-6/$SiO_2$ increased (up to 2 wt%), the performance differences in terms of voltage and current also grew. At a ratio of 2 wt%, the output difference between KIT-6 and $SiO_2$ was at its maximum. Combining these results, we can draw several conclusions. Firstly, when the same amount of KIT-6 was introduced into flat PDMS and porous PDMS, the difference in performance was minimal. However, when comparing KIT-6 and non-porous $SiO_2$ in porous PDMS, the output difference was significantly greater, highlighting the significant role of the unique properties of $SiO_2$ its large SSA and uniform mesoporous structure in enhancing TENG performance. Initially, the TENG performance saw a notable enhancement attributed to the charge retention ability of the mesoporous KIT-6 nanoparticles. KIT-6 features large, ordered domains with a pure bicontinuous mesostructure. This well-defined structure is essential for improving TENG performance because it provides a substantial surface area for charge storage and efficient pathways for charge transfer [18]. Furthermore, the high SSA of KIT-6 plays an important role in improving TENG output performance. The unique 3D bicontinuous pore structure of KIT-6 also effectively enhances performance. This structure enhances

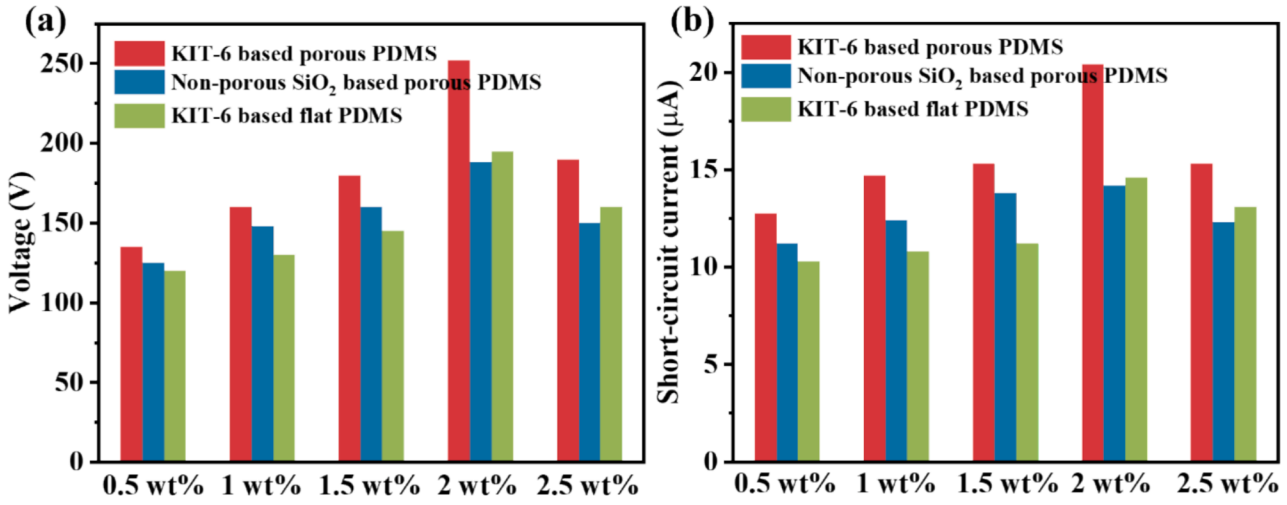



**Fig. 5.** (a) Comparison of output voltage and (b) current.

**Table 1**
A summary of key output voltage and current data.

| | 0.5 wt% | | 1 wt% | | 1.5 wt% | | 2 wt% | | 2.5 wt% | |
|---|---|---|---|---|---|---|---|---|---|---|
| | Voltage (V) | Current (μA) | Voltage (V) | Current (μA) | Voltage (V) | Current (μA) | Voltage (V) | Current (μA) | Voltage (V) | Current (μA) |
| A | 135 | 12.8 | 160 | 14.7 | 180 | 15.3 | 252 | 20.4 | 190 | 15.3 |
| B | 125 | 11 | 148 | 12.4 | 160 | 13.8 | 188 | 14.2 | 150 | 12.3 |
| C | 120 | 10 | 130 | 10.7 | 145 | 11.2 | 195 | 14.6 | 160 | 13.1 |

A: KIT-6 based porous PDMS; B: Non-porous $SiO_2$ based porous PDMS; KIT-6 based flat PDMS.

charge transport and retention by creating pathways for electron flow that do not exist in non-porous $SiO_2$. Additionally, the interconnected pores of KIT-6 allow for a greater volume of confined air and surface interactions, contributing to an enhanced triboelectric effect.

The nanogenerator's performance was evaluated across varying working loads and frequencies to simulate practical conditions where environmental stimuli, such as vibrations, often fluctuate. Fig. 6a–c presents the voltage output, short-circuit current, and charge transfer density of the nanogenerator under varying loads (1–7 N). At 1 N, the TENG achieved an output voltage of 87 V, short-circuit current of 7.7 μA, and a charge transfer density of 34 μC/m$^2$, which were slightly higher than those observed in the device using pure PDMS at 5 N. This initial enhanced performance is linked to the rougher surface created by the inclusion of KIT-6 nanoparticles. As the applied force rose from 1 N to 7 N, the TENG's voltage improved notably, primarily because of the expanded effective contact area. The maximum contact area was reached at a load of 6 N, after which further load increases to 7 N resulted in only minimal output enhancement. Additionally, as the triboelectric layers of porous PDMS and copper separated, charge accumulation occurred until it eventually stabilized. At a load of 7 N, the nanogenerator showed an output voltage of 316 V, a short-circuit current of 24.6 μA, and a charge transfer density of 73 μC/m$^2$. Fig. 6d–f displays the TENG's performance across various frequencies. Increasing the motion speed from 0.5 to 3 Hz elevated the output voltage from 84 V to 301 V and the short-circuit current from 7.8 μA to 25.4 μA, as illustrated in Fig. 6d. Both the output voltage and short-circuit current linearly increased with frequency. However, the charge transfer density remained constant because the effective contact area remains stable at a given load regardless of frequency. The observed rise in charge transfer rate with higher frequency is attributed to the accelerated rate of charge movement between the electrodes.

A resistor box ranging from 0.2 MΩ to 900 MΩ was used to appley external resistances. Fig. 7a, b displays the short-circuit current and open-circuit voltage measured as functions of external resistance, with a moving speed of 2 Hz and an applying force of 5 N. In Fig. 7a, the output voltage rose gradually, peaking at 254 V, as the external load resistance increased. The open-circuit voltage leveled off at approximately 250 V once the resistance went beyond 300 MΩ, which was determined to be the TENG's open-circuit voltage. Conversely, the short-circuit current diminished with higher load resistance, as shown in Fig. 7b. Moreover, the maximum output current, 20.4 μA, was observed at an external resistance of 0.2 MΩ. As shown in Fig. 7c, the power density initially increased rapidly and then decreased after reaching its peak value. The maximum power density achieved by the KIT-6/porous PDMS TENG was approximately 4.37 W/m$^2$ at 20 MΩ. This output significantly surpasses that of the unmodified PDMS TENG, which achieved a maximum power density of 0.67 W/m$^2$, and the porous PDMS TENG, which has that of 0.34 W/m$^2$. This improvement underscores the significant impact of mesoporous KIT-6 on enhancing the power generation capabilities of the nanogenerator. The KIT-6/porous PDMS TENG

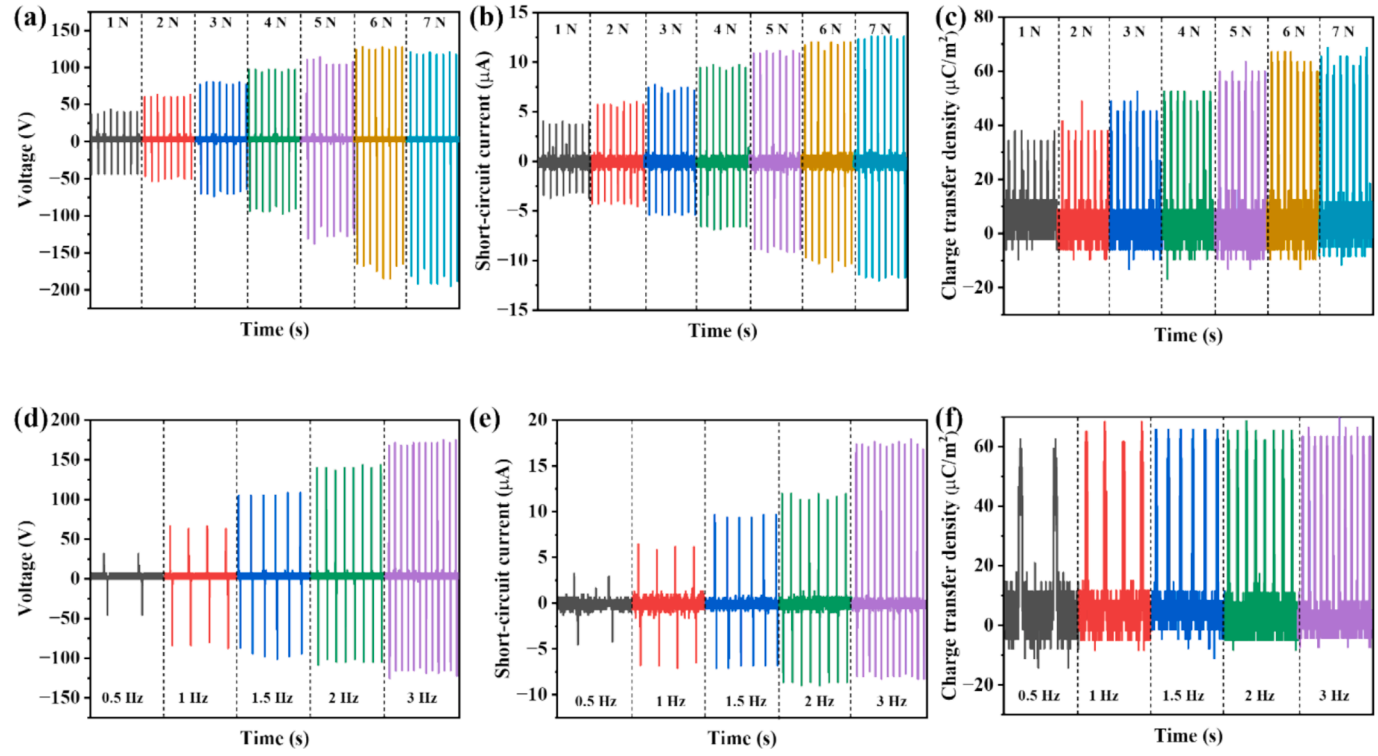



**Fig. 6.** (a) Voltage, (b) current, and (c) charge transfer density of the TENG with KIT-6 were measured under varied working loads. ranging from 1 to 7 N. Additionally, the (e) voltage, (f) current, and (g) charge transfer density of the TENG with KIT-6 were evaluated at varying frequencies, from 0.5 to 3 Hz.

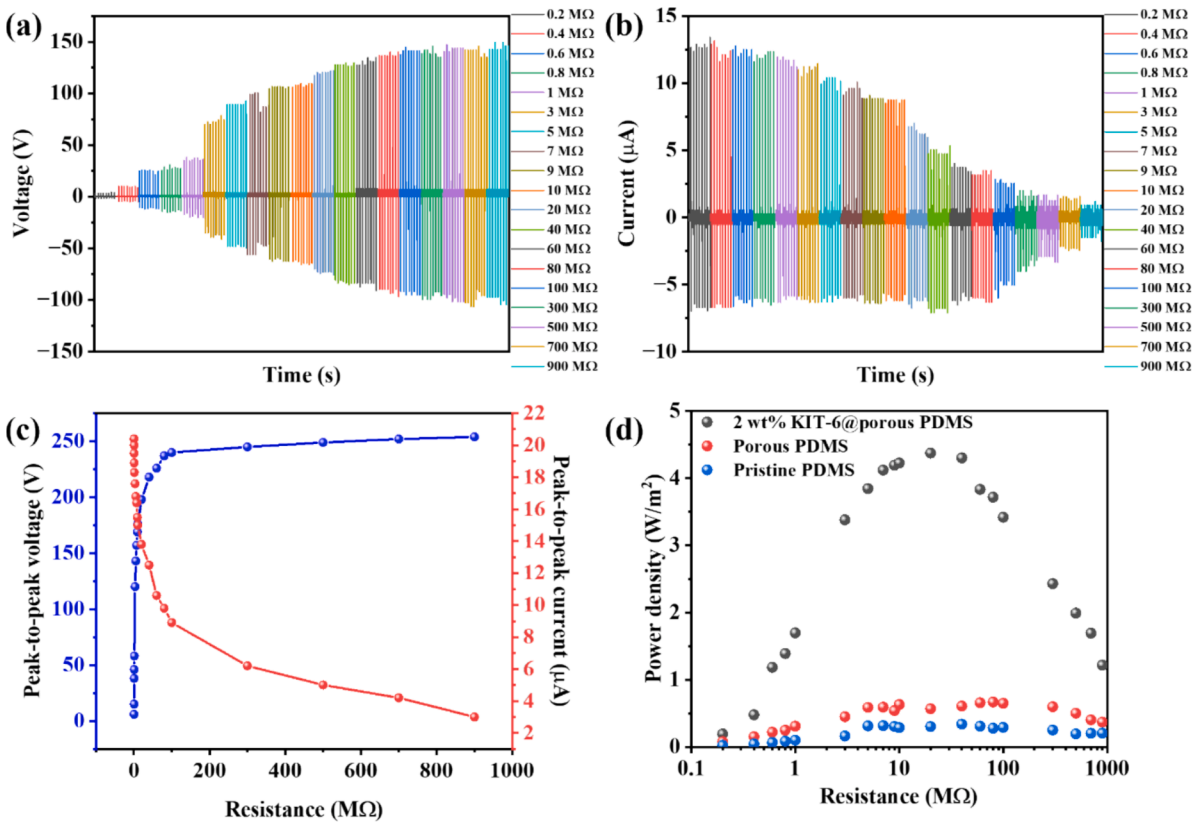



**Fig. 7.** Power density evaluation with external load resistance: (a) Voltage and (b) current as functions of load resistance. (c) The relationship between peak-to-peak voltage and current at varied external load resistances. (d) Maximum power density as a function of external load resistance.

demonstrated superior output performance compared to both pristine PDMS TENG and porous PDMS TENG, highlighting its ability to enhance energy harvesting and its real-world uses in self-powered sensors, wearable technology, and a variety of energy harvesting systems. In other words, our findings highlight that the incorporation of KIT-6 into porous PDMS significantly improves the triboelectric performance of TENGs. The KIT-6′s mesoporous structure, characterized by its extensive surface area, serves effectively as a charge storage site, leading to enhanced overall charge density and improved power output. This research provides important insights into the design and optimization of friction layer materials, advancing the development of sustainable energy technologies.

As shown in Fig. 8, the potential of the KIT-6/porous PDMS TENG for self-powered energy harvesting was evaluated. The electrical power produced was accumulated inside a capacitor through a rectification circuit, depicted in Fig. 8a. The KIT-6/porous PDMS TENG's output

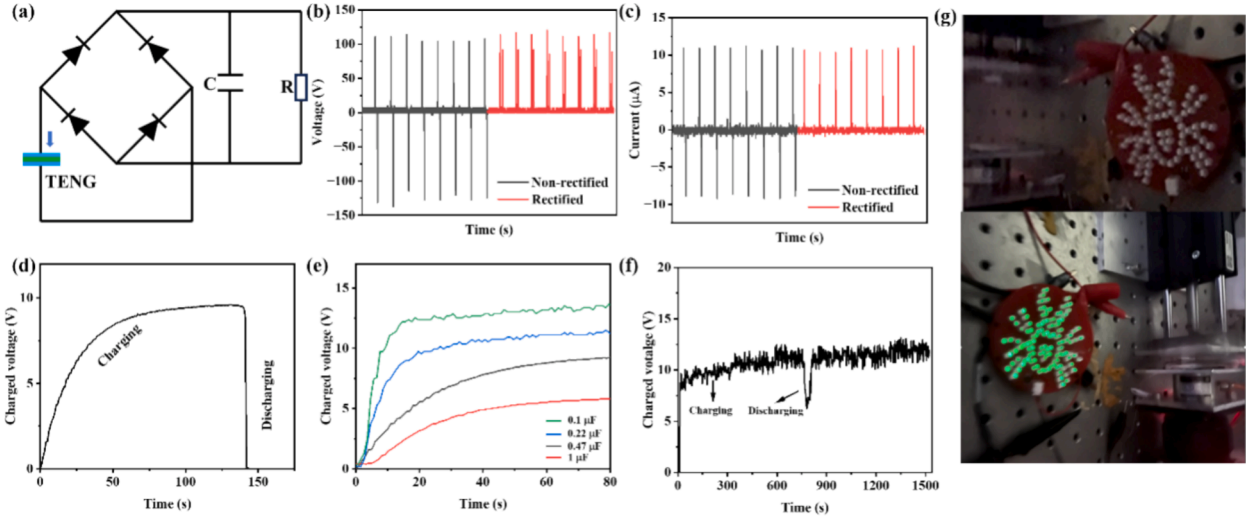



**Fig. 8.** Self-sustaining energy generator using PDMS-based TENG infused with KIT-6: (a) Schematic illustration of the rectifier circuit with a capacitor. (b) Output voltage measured with a rectifier. (c) Rectified output current measured with a rectifier. (d) Capacitor charging and discharging characteristics. (e) Charging performance of capacitors with different capacitances (0.1 μF, 0.22 μF, 0.47 μF, and 1 μF). (f) Charging curve of the capacitor (2 Hz and 5 N). (g) Illustration showing the illumination of commercial LEDs powered by the KIT-6/porous PDMS-based TENG.

performances were converted using a full bridge rectifier, illustrated in Fig. 8b, c. The rectified voltage and current values were similar to the original unrectified measurements, indicating that the device's output is able to be effectively transformed into a rectified curve with minimal loss. Fig. 8d shows the charging and discharging characteristics of the circuit with a capacitor. The capacitor (0.1 μF) charges quickly, with a rapid rise in voltage. The sudden drop observed in the graph corresponds to the discharge phase when the KIT-6/porous PDMS TENG was deactivated. The charging characteristics of capacitors with varying capacitances (0.1 μF, 0.22 μF, 0.47 μF, and 1 μF) exhibited patterns similar to those described in earlier studies, showing an initial rapid increase in voltage followed by a more gradual charging process, as illustrated in Fig. 8e. Capacitors with larger capacitances required more time to reach a given voltage compared to those with smaller capacitances. Fig. 8f illustrates that the capacitor was able to be recharged when reconnected to the KIT-6/porous PDMS TENG, demonstrating the device's capability to effectively store energy in capacitors. This study highlights the KIT-6/porous PDMS TENG's capability to effectively harvest and store energy, making it a viable option for powering small electronic devices and contributing to sustainable energy solutions. The results highlight the critical role of friction material optimization in improving the performance and efficiency of TENGs.

In this work, the KIT-6/porous PDMS TENG created was utilized to directly operate compact electronic devices for various applications. As shown in Fig. 8g, the TENG successfully lit 84 commercial LEDs in series. The KIT-6/porous PDMS TENG was affixed to clothing and insoles to evaluate its application as a sensor. Fig. S4a depicts the nanogenerator attached to a slipper insole, demonstrating its capability to produce a peak output voltage of 4.5 V as the sole continuously contacts and separates from the nanogenerating device. Fig. S4b illustrates the TENG attached to a leg, creating a peak open-circuit voltage of 1.5 V during movements like leg clapping and separation. Fig. S4c shows the nanogenerator mounted on a coat generating a maximum output voltage of around 2.9 V. This lower voltage is attributed to the different triboelectric properties of fabric compared to human skin. In Fig. S4d, the TENG affixed to waist clothing produced a maximum output voltage of approximately 3.3 V when the elbow swung during running. The waveform distinctly captured the phases of contact and separation. These results demonstrate that the output voltage waveforms vary with different body movements, suggesting the potential of TENGs as sensors to deduce human actions based on the corresponding waveforms. This innovative TENG design using KIT-6 in combination with porous PDMS demonstrates improved performance and stability. It represents a positive solution for sustainable energy applications, including self-powered sensors, wearable devices and energy harvesting systems. The ability to light 84 LEDs signifies the potential of KIT-6/porous PDMS TENGs for practical energy harvesting and utilization. KIT-6 offers promising properties for TENG performance but faces scalability challenges due to time-consuming and costly templating steps. Potential solutions include the use of cheaper templates and the development of faster synthesis methods.

## 4. Discussions

In summary, we fabricated a nanogenerator using a porous PDMS structure integrated with mesoporous KIT-6 to significantly enhance its output performance. The optimized KIT-6/porous PDMS TENG achieved impressive values of 254 V for output voltage, 20.4 μA for short-circuit current, and 70.8 $\mu C/m^2$ for charge transfer density at a 2 wt% KIT-6 concentration, which represent improvements of 3.0, 2.7, and 2.1 times, respectively, over a pristine PDMS TENG. Excessive KIT-6 concentration resulted in reduced benefits due to reduced effective PDMS area as the nanoparticles aggregated on the surface. Performance improved significantly with increasing loading frequency and applied load, achieving a peak power density of 4.37 $W/m^2$ under ideal conditions. Practical applications were demonstrated by attaching the TENG to pants and waist, where it effectively harvested mechanical energy from various movements. In addition, the TENG successfully powered a series of 84 green LED bulbs. This study offers important perspectives on enhancing the nanogenerator's performance and highlights its potential for commercial applications.

## CRediT authorship contribution statement

**Jianbin Mao:** Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yutao Wen:** Writing – original draft, Methodology, Investigation, Data curation. **Xiguang Feng:** Writing – original draft, Investigation, Data curation. **Weiming Xu:** Writing – original draft, Supervision, Data curation. **Soonmin Seo:** Writing – review & editing, Writing – original draft, Supervision, Funding acquisition, Data curation, Conceptualization.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.apsusc.2025.162291.

## Data availability

Data will be made available on request.

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